

The Aharonov–Bohm Effect

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Key Experiments in Quantum Science Seminar

July 2, 2026

The Aharonov–Bohm effect shows that the observable quantum phase can depend on global properties of electromagnetic potentials even though the particle trajectories lie in a field-free region. This paper explains the effect and discusses the experiments by Chambers and Tonomura. Particular emphasis is placed on Tonomura’s improvements and on the role of electron holography, which enabled an unambiguous confirmation of the magnetic Aharonov–Bohm effect.

1 Introduction

As early as 1949, Werner Ehrenberg and Raymond E. Siday described a magnetic phase shift in electron optics.¹ Yakir Aharonov and David Bohm independently developed this idea further at the University of Bristol in 1959. They showed that charged particles can acquire a relative phase even when they travel only through regions in which the electromagnetic fields vanish.²

The effect now named after Aharonov and Bohm sparked intense debate in the scientific community. In classical electrodynamics, potentials can often be regarded as auxiliary quantities used to calculate the fields; in quantum mechanics, however, their line integral contains globally measurable information. What is observable is not the gauge-dependent potential at a single point, but the gauge-invariant phase difference between different paths.³

Chen Ning Yang, Nobel laureate and co-founder of modern gauge theories, studied the AB effect extensively. Together with Tai Tsun Wu, he identified the nonintegrable phase factor as a fundamental geometric object in gauge theories.⁴

This paper first explains the theoretical idea behind the effect and discusses the earliest experimental test and its limitations. The main part then focuses on Akira Tonomura’s experiment, which convincingly confirmed the magnetic effect with completely shielded flux. Finally, the current state of the field and open prospects for the AB effect are outlined.

2 Theoretical Foundations

2.1 Phase Factor

The starting point for the theoretical discussion is the Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = \hat{H} \psi(\mathbf{x}, t) \quad (1)$$

Consider a charged particle in an electric potential

$\Phi(t)$ that is spatially constant but varies with time. Classically, its potential energy is given by $V(t) = q\Phi(t)$. After canonical quantization, the Hamiltonian is

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + q\Phi(t) \quad (2)$$

Because the potential is spatially constant, the term $q\Phi(t)$ acts only as a time-dependent scalar. It therefore commutes with the kinetic part of the Hamiltonian. The two contributions affect the time evolution independently and may thus be considered separately. For the potential term alone, one obtains

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = q\Phi(t) \psi(\mathbf{x}, t) \quad (3)$$

This differential equation can be solved by separation of variables, yielding

$$\psi(\mathbf{x}, t) = \exp\left(-\frac{iq}{\hbar} \int_0^t \Phi(t') dt'\right) \psi(\mathbf{x}, 0) \quad (4)$$

The electric potential therefore contributes only a phase factor to the time evolution of the wave function. Such a global phase is not directly observable on its own. It becomes observable, however, when two coherent partial waves acquire different phases and are subsequently recombined to interfere. For two paths subject to different potentials, for example, the relative phase shift is

$$\Delta\varphi = -\frac{q}{\hbar} \int_{t_i}^{t_f} (\Phi_1(t) - \Phi_2(t)) dt \quad (5)$$

This phase can be generalized to the electromagnetic case. Aharonov and Bohm derived this generalization using the path-integral formulation of quantum mechanics.² The time evolution of a charged particle along a path is then described by a path-dependent phase factor:

$$\exp\left[\frac{iq}{\hbar} \left(\int \mathbf{A} \cdot d\mathbf{r} - \int \Phi dt\right)\right] \quad (6)$$

For two paths that together form a closed curve C , Stokes' theorem gives the gauge-invariant magnetic phase difference

$$\Delta\varphi_{AB} = \frac{q}{\hbar} \oint_C \mathbf{A} \cdot d\mathbf{r} = \frac{q}{\hbar} \int_S \mathbf{B} \cdot d\mathbf{S} = \frac{q}{\hbar} \Phi_B. \quad (7)$$

The measurable phase thus depends only on the enclosed magnetic flux Φ_B . It is the central quantity in the Aharonov–Bohm effect.⁴ Its consequences are now illustrated using two idealized experiments.

2.2 AB Experiments

Aharonov and Bohm originally proposed two experiments to demonstrate the physical significance of their path-dependent phase factor, or equivalently of the scalar and vector potentials.² These are now known as the electric and magnetic AB effects. In the electric setup, a coherent electron wave packet is split into two partial beams that pass through separate conducting tubes. While the electrons are inside the tubes, the tubes are raised to different electrostatic potentials. Before the electrons reach the ends of the tubes, the potentials are returned to zero. The two partial beams are then recombined so that the expected phase shift can be measured.

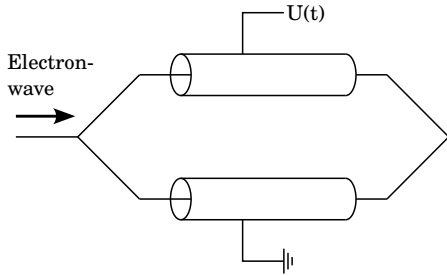


Fig. 1: Setup of the electric AB effect.

In the idealized description using infinitely long tubes, the electrons are not exposed to an electric field; nevertheless, a relative phase shift arises from the time-integrated potential difference. The magnetic AB effect is analogous: two coherent electron beams pass on opposite sides of a shielded solenoid, idealized as infinitely long, and are subsequently recombined to interfere.

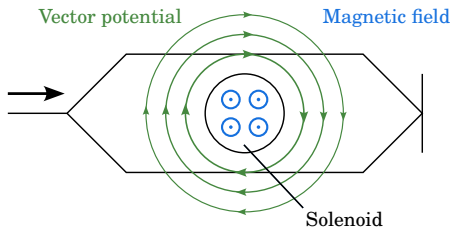


Fig. 2: Setup of the magnetic AB effect.

The magnetic field is completely confined to the interior of the solenoid. The electrons therefore again travel through a field-free region in which the vector potential, or the associated phase factor, is nevertheless nontrivial. The phase formula consequently predicts a phase

difference between the electron beams. This difference is proportional to the enclosed magnetic flux.

Both cases rely on idealizations. The tubes or solenoid are assumed to be sufficiently long, and the electromagnetic fields are shielded or localized so that they do not reach the electrons along their paths. The central claim of both experiments is therefore that electrons can acquire a measurable phase shift even though they travel through a region in which the electromagnetic fields vanish. A local and fundamental description of the interaction consequently cannot rely on the local field strengths alone.

3 Chambers' AB Experiment (1960)

3.1 Setup

Because the experimental realization of the electric AB effect requires time-dependent potentials, early research concentrated on the magnetic variant. R. G. Chambers performed the first experimental test in 1960, only one year after the theoretical publication by Aharonov and Bohm.⁵ Chambers used a modified electron microscope as a source of coherent electrons. To produce the interference required for the effect, the electron beam was split using an electrostatic biprism. The biprism consisted of an aluminized quartz filament approximately 1.5 μm in diameter. The filament divided the beam and cast a geometric shadow, while a positive voltage deflected the two partial waves electrostatically toward one another. In the overlap region behind the filament, a pattern of approximately equally spaced interference fringes formed on a photographic film.⁵

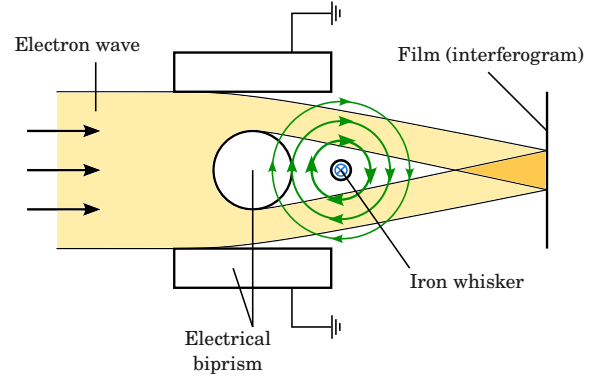


Fig. 3: Setup of Chambers' experiment.

Instead of a solenoid, Chambers placed a magnetized iron whisker in the geometric shadow of the biprism filament. The single-crystal iron needle was approximately 1 μm wide and 0.5 mm long, consisted of a single magnetic domain, and tapered to a point.^{5;6}

3.2 Results

The results show characteristically tilted interference fringes, while the outer boundary of the pattern remained stationary. The explanation of this tilt leads to the main criticism of the experiment. Owing to the

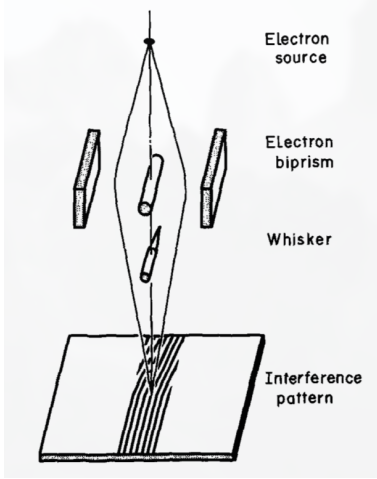


Fig. 4: Three-dimensional setup of Chambers' experiment.⁶

conical geometry, the magnetic flux enclosed by the two partial beams decreased along the whisker, continuously shifting the interference pattern. Critics pointed out, however, that stray fields could emerge from the tapered whisker and deflect the partial beams through the Lorentz force. A purely force-free explanation of the fringe tilt was therefore not yet established beyond doubt.⁷

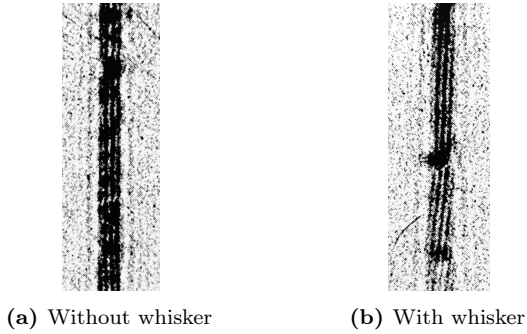


Fig. 5: Comparison of the interference patterns.

Numerous variants of the AB experiment were realized over the following decades. Tonomura's group achieved a particularly convincing confirmation in 1986 by strictly separating the electron wave from the magnetic field.^{6;7}

4 Tonomura's AB Experiment (1986)

Akira Tonomura worked at the Advanced Research Laboratory of Hitachi Ltd. His group made decisive advances there in field-emission electron microscopy and electron holography. On this basis, the group designed an AB experiment that was carried out in 1986 by a large collaboration and confirmed the magnetic effect with completely shielded flux.^{6;8}

The improvements over earlier experiments were based primarily on three elements: an improved electron source, electron holography as a new interferometric technique, and new methods for separating the electron wave from the magnetic field. These elements are first discussed individually, followed by a description of the

complete setup.

4.1 Electron Source

A field-emission electron microscope developed at Hitachi served as the source of coherent electron waves. The central component of the field-emission gun was a tungsten tip with a radius of approximately 100 Å. Unlike the thermal sources commonly used at the time, electrons were extracted by tunneling through the strong electric field between the tip and the first anode. A second anode accelerated them to 150 kV. To stabilize and protect the tip, the source was operated at a pressure on the order of 1×10^{-10} Torr. Condenser lenses then formed an almost parallel beam.⁶

The field-emission source provided substantially greater spatial coherence and approximately an order of magnitude better resolution than the apparatus used by Chambers. With a related setup, Tonomura's group later also demonstrated the gradual buildup of an interference pattern from individually detected electrons.^{6;9}

4.2 Field Shielding

The criticism of earlier experiments ultimately centered on the insufficient separation of the electron wave from the magnetic field. Although Chambers' open whisker geometry facilitated the interference measurement, it allowed stray fields. Tonomura's group therefore used a microscopic, lithographically fabricated ring made of a ferromagnetic alloy. Its toroidal geometry closes the magnetic field lines and thus avoids the problematic open ends of the whisker. The use of a torus for the AB effect was proposed by G. T. Trammel.^{6;10}

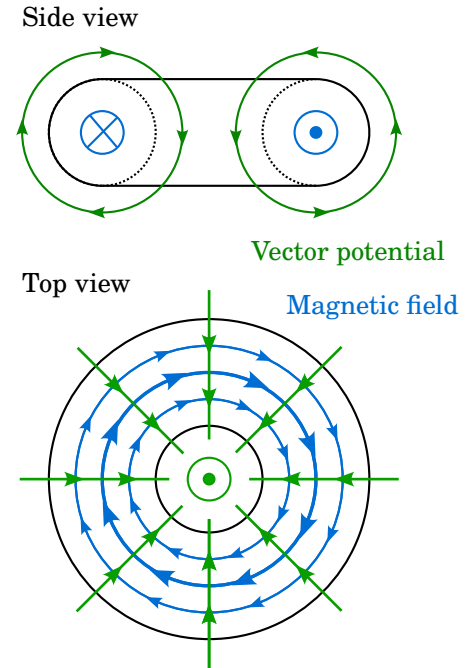


Fig. 6: Magnetic field of a toroidal ring.

The phase of partial waves passing through the central opening is compared with that of partial waves passing

outside the torus.

For additional shielding, the ring was coated with a 2500 Å-thick niobium layer and cooled below the transition temperature during the experiment. The Meissner–Ochsenfeld effect prevented magnetic flux from penetrating the superconducting layer and reaching the electron paths. A copper layer between 500 Å and 2000 Å thick also blocked the electron wave inside the material and reduced electrostatic charging. The finished sample was mounted on a microscopic copper grid.^{6;8}

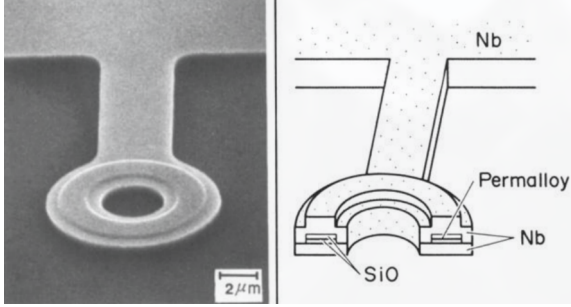


Fig. 7: Toroidal sample in the AB setup.⁸

4.3 Electron Holography

Electron holography is a two-stage imaging technique. Whereas photography records only the intensity of a wave, a hologram uses interference with a reference wave to encode its relative phase as well. In the second stage, the recorded wavefront can be restored using a suitable reconstruction wave. The principle relevant to Tonomura’s experiment is explained below using a simple example.^{6;11}

Suppose that a laser beam is split into two parts. One part strikes an object and is scattered as the object wave. The second part passes the object as nearly unchanged as possible and forms the reference wave. The two waves are then superposed on a photographic plate or film.

The film does not measure the phase of the light wave directly, but only its intensity. The phase information of the object wave alone would therefore be lost. When the object and reference waves are superposed, however, they produce an interference pattern whose fringe positions depend on their relative phase. The phase is thus not photographed directly, but is stored in the positions of the interference fringes.

Mathematically, this can be expressed concisely: the reference and object waves are superposed on the film, so the measured intensity is proportional to

$$I = |R + O|^2 = |R|^2 + |O|^2 + R^*O + RO^* \quad (8)$$

This expression contains not only the individual intensities of the reference and object waves but also cross terms. These contain the relative phase between the two waves. A hologram therefore contains more information than an ordinary photograph.

The simplest case is the hologram of a single point. A single object point emits approximately a spherical

wave, while the reference wave is planar. Their superposition produces a system of rings marking positions of constructive and destructive interference. The hologram therefore does not resemble the object directly; instead, it is an encoded record of its wavefront.

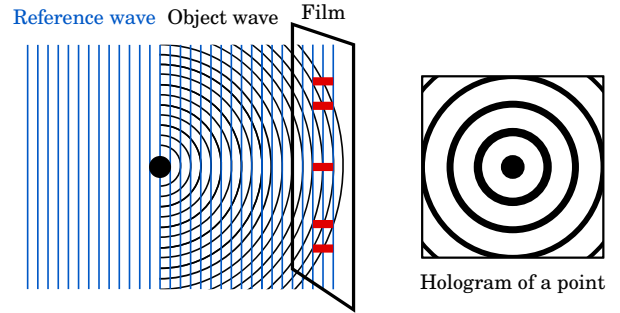


Fig. 8: Recording a point hologram.

During reconstruction, the developed hologram is illuminated by a wave corresponding to the original reference wave. Diffraction by the hologram again produces a wave with the form of the original object wave. For a single object point, this creates another spherical wave that appears to originate at the point’s original position.

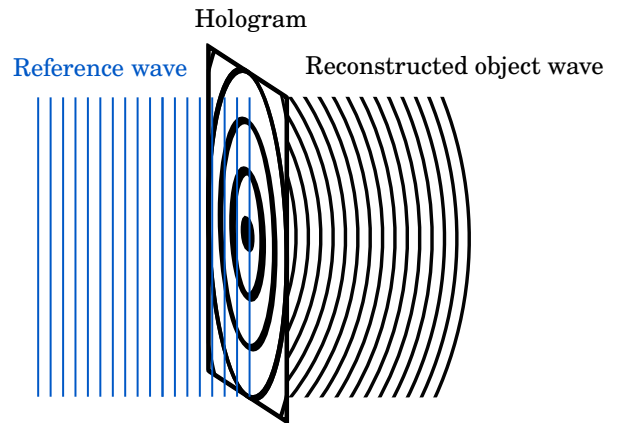


Fig. 9: Reconstruction of a point hologram.

The point example can be extended to an object of finite size. Each object point contributes its own partial wave and therefore its own interference pattern; the complete hologram is the superposition of all these patterns.

Because wave propagation is linear, the amplitudes of the individual point waves add. Reconstruction therefore restores the complete object wavefront, including its phase information.

This principle allows the phase information contained in the electron beam to be stored first in a hologram and subsequently read out optically.

4.4 Setup

To record the electron hologram, the torus was placed in one half of the coherent electron beam. This half formed the object wave, while the other half formed the reference wave. Within the object wave, partial

waves passed either through the opening or outside the torus; electrons striking the ring were absorbed. As in Chambers' experiment, a Möllenstedt biprism deflected the object and reference waves toward one another and caused them to interfere. Before the pattern was recorded on film, an electron lens magnified it.⁶

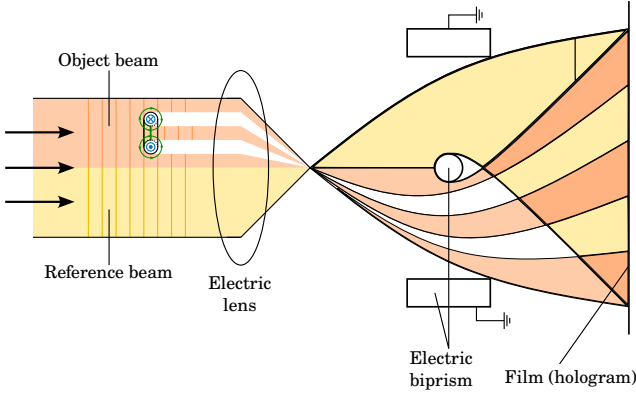


Fig. 10: Setup of Tonomura's electron holography experiment.

Reconstruction was performed not with electrons, but with light from a He–Ne laser. Magnification was necessary because the electron wavelength was approximately $0.03 \text{ \AA}$, whereas the laser wavelength was $6328 \text{ \AA}$. For a geometrically faithful angular reconstruction, a magnification of approximately 210 000 would have been optimal; the experimentally achieved magnification of 1000 was sufficient to detect the phase.⁶

Illuminating the hologram with the laser reconstructed the original electron wavefront, including its phase structure, as an optical wave. An ordinary camera would nevertheless record only the intensity and therefore essentially the shadow of the torus. To make the phase visible, the reconstructed wave was consequently superposed with another reference wave in a Mach–Zehnder interferometer.

For this purpose, the He–Ne laser was split into two beams. After one beam had passed through the hologram, the reconstruction and reference beams were superposed at the camera; an additional holographic technique doubled the fringe density. This completed the electron-holographic process.⁶

After this discussion of the experimental setup, the results are considered next.

4.5 Results

The final interferogram shows a clear phase shift between the partial waves passing inside and outside the torus. Its magnitude agrees with the theoretical prediction of Aharonov and Bohm within the experimental accuracy. Because the magnetic flux was separated from the electron wave by both the toroidal geometry and the superconducting niobium layer, the experiment is regarded as an unambiguous confirmation of the magnetic AB effect.⁸

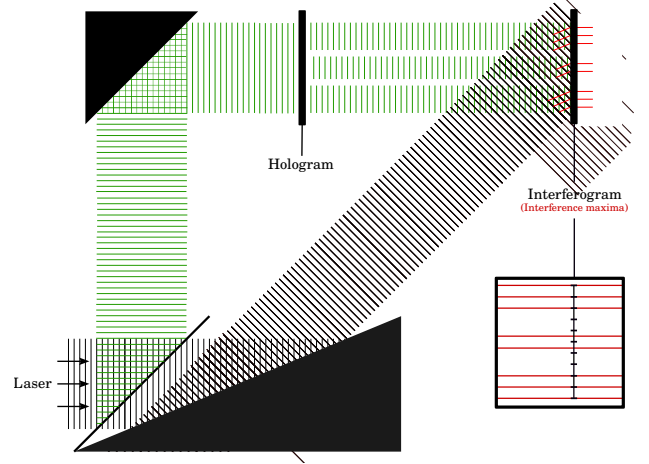


Fig. 11: Conversion of the hologram into an interferogram.

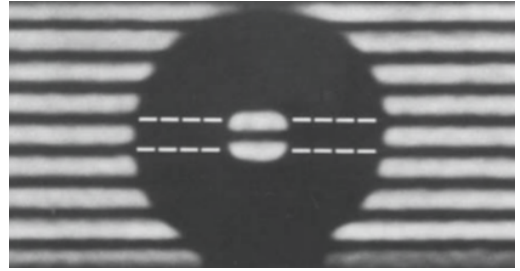


Fig. 12: Final interferogram of the experiment.⁸

5 Significance and Outlook

Finally, the significance of the effect and the current state of research are briefly assessed.

Electron holography and electron holographic microscopes are versatile tools in experimental physics and can, for example, image microscopic electric and magnetic fields quantitatively. An incorrectly fabricated sample from Tonomura's experiment provides one example: the position-dependent displacement of the interference fringes reveals leaking magnetic flux and enables its quantitative determination.⁶

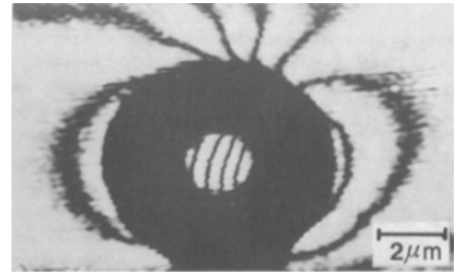


Fig. 13: Interferogram of a defective torus.⁸

Conceptually, the effect shows that the local quantities $\mathbf{E}$ and $\mathbf{B}$ alone can miss global, topological information. Electrodynamics is an Abelian $U(1)$ gauge theory; in 1954, Yang and Mills generalized the gauge principle to non-Abelian groups. For the AB effect, the holonomy of the $U(1)$ connection is decisive: the potential is gauge dependent, whereas its closed-loop phase factor is observable.^{3;4}

The original time-dependent electric AB effect has still not been demonstrated unambiguously. Its realization is made difficult primarily by the rapid potential switching, field shielding, and required coherence. In addition to ion interferometers, energy shifts of shielded quantum systems have recently been proposed as an experimental approach.^{12;13}

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